Machine Learning Assignment 3

Topics: Decision Tree

Total Marks: 10

Note: You can use inbuilt scikit learn implementation for the experiments.

Q1. Classification: Consider the Car Evaluation dataset (available at `https://archive.ics.uci.edu/dataset/19/car+evaluation`) and implement the following:

- Grow a decision tree using either the entropy measure (ID3 algorithm) or the Gini index (CART algorithm). Decision trees tend to be overly complex and may not generalize well to unseen data, leading to overfitting. To address this, use stopping criteria such as limiting the maximum depth of the tree T.

  Use cross-validation to determine the optimal depth T that maximizes performance. Plot the average accuracy as a function of T. Visualize the decision tree constructed on the training data.

- **Ensemble methods to overcome overfitting:**
  - **Bagging:** For the optimal value of T identified above, perform bagging using B bootstrapped datasets. Consider B in the range of 50 to 500 with increments of 50. Use cross-validation to select the optimal value of B. Plot the average accuracy as a function of B.
  - **Random Forest:** Construct a random forest consisting of B trees using the optimal value of B obtained above. For each tree, at each split, randomly select m out of d features (where m can range from $\sqrt{d}$ to d). Compute the average classification accuracy for $m = 4$.

Q2. Regression: Using the Boston Housing dataset (available at `http://lib.stat.cmu.edu/datasets/boston`), perform decision tree regression to predict the value of `MEDV`, which represents the median price of owner-occupied homes.

- Grow a regression tree using recursive binary splitting. Decision trees tend to be overly complex and may not generalize well to unseen data, leading to overfitting. To address this, use stopping criteria such as limiting the maximum depth of the tree or the minimum number of samples in a terminal node S. Use $S = 4$.

  Visualize the regression tree constructed on the training data.

- **Ensemble methods to overcome overfitting:**
  - **Bagging:** For the value of S identified above, perform bagging using B bootstrapped datasets. Consider B in the range of 50 to 500 with increments of 50. Use cross-validation to determine the optimal value of B. Plot the average mean squared error (MSE) as a function of B.
  - **Random Forest:** Construct a random forest consisting of B trees using the optimal value of B obtained above. For each tree, at each split, randomly select m out of d features (where m ranges from $\sqrt{d}$ to d). Compute the average prediction accuracy for $m = 7$.